EcoLoop: Real Builder Recording Script

Natural, direct, peer-to-peer read-aloud script. Zero corporate fluff.

Harman • SSDPS Jalandhar
Target: 12 to 15 Minutes

> **Delivery Vibe:** Talk like you're on a Discord call or talking to other makers your age. Calm, deadpan, grounded. No fake excitement. No preaching about saving the planet. You built a real physical product and real software with $0, and you're just showing how you pulled it off.

SLIDE 01 • TITLE

EcoLoop: Circular material systems and zero-cost digital architecture

"Hey everyone. I'm Harman, from Swami Sant Dass Public School in Jalandhar, India."

"Look, most school environmental projects are kind of a joke. People make a poster, pick up three plastic bottles after lunch, take a photo for Instagram, and call it sustainability. We wanted to actually build real things instead."

"So EcoLoop has two parts: first, taking trash that recyclers literally throw away and turning it into solid, usable materials like structural boards. And second, building a digital platform to track and distribute those products for zero dollars in running costs."

"Let's get into the material problem first." `[NEXT SLIDE]`

SLIDE 02 • THE PROBLEM

The Linear Packaging Problem

"When people talk about recycling, everyone obsesses over plastic water bottles. But bottles are actually easy to melt down. The real nightmare is multi-layer packaging, or MLP: chips packets, chocolate wrappers, biscuit packs."

"They take shiny aluminum foil and laminate it between plastic layers. Because those layers are chemically bonded together, municipal recycling plants don't touch them. It costs more money to separate the foil than the plastic is worth. So virtually 100% of snack packaging ends up burned or buried in a dump."

"Meanwhile, every school canteen generates sacks of clean wrappers every single day. So our solution wasn't separating the layers; it was heat-compressing the whole thing together into heavy structural sheets that can replace commercial plywood." `[NEXT SLIDE]`

`SLIDE 03 • THE PHYSICAL LOOP`

The Physical Loop: Verified Materials

"These are the six physical products we actually produce."

"On the right, you can see an actual sample of our MLP Sheet. That board is made entirely out of compressed snack packets. You can drill it, saw it, hammer nails into it, and it's completely waterproof and termite-proof. It basically behaves like synthetic lumber."

"We also make Eco-Bricks by packing clean plastic bags into bottles for construction, Bottle Planters for gardening, Uniform Bags stitched from old school uniforms thrown away at the end of the year, handmade Seed Paper with plantable seeds, and Bioenzymes fermented from fruit peels for cleaning. Every product takes a specific waste stream and turns it into something you can actually use." `[NEXT SLIDE]`

`SLIDE 04 • REALITY CHECK 01`

Reality Check: How It Started vs How It Is Going

"Now, a quick reality check on how this actually went down. **(deadpan)**"

"When we first pitched this, the teachers thought we were just going to set up blue and green bins around school and pick up empty Lay's packets after break."

"Within two weeks, we realized that handling orders, publishing material density specs, and keeping track of inventory meant we needed an entire software platform. So we ended up staying up late writing code instead of just picking up trash. Let's look at the tech." `[NEXT SLIDE]`

`SLIDE 05 • DIGITAL LAYER`

Why Physical Projects Need Software

"Why does a physical recycling project need software?"

"Because if you just say 'hey, buy my recycled plastic board', nobody takes you seriously. You have to give people actual engineering numbers: thickness, density, thermal ratings, and delivery times. That's what turns a school science project into real materials people can trust."

"On our web platform, partners can browse the catalog, inspect physical specs, and request samples directly. We built the frontend using React 18, TypeScript, and Vite so it loads instantly and doesn't break at runtime." `[NEXT SLIDE]`

`SLIDE 06 • SERVERLESS BACKEND`

The Zero-Cost Serverless Backend

"Here was the main constraint: we had zero dollars. Literally zero."

"We're high school students. None of us have a company credit card to pay $50 every month for AWS servers, Docker instances, or database hosting. So we built a completely free serverless setup:"

"Step 1: The user fills out an inquiry form in React, and the client checks the inputs. Step 2: The data hits a Google Apps Script webhook in the cloud. Step 3: That webhook dumps the order straight into a Google Sheet, which acts as our live database and pings our email instantly. **(pause)** Total monthly server bill: zero dollars and zero cents. It's fast, free, and has 99.9% uptime." `[NEXT SLIDE]`

`SLIDE 07 • REALITY CHECK 02`

Reality Check: The Cloud Bill Dilemma

"Which brings us to reality check number two."

"You go on Twitter or LinkedIn and see enterprise software engineers arguing for six months about Kubernetes clusters and microservices, burning hundreds of dollars a month on cloud hosting."

"Meanwhile, two high school kids run their entire production order backend on a free Google Sheet, and it literally never goes down." `[NEXT SLIDE]`

`SLIDE 08 • INQUIRY ENGINE`

Interactive Inquiry Engine

"Here's what that inquiry flow looks like in the app."

"When someone clicks 'Enquire' on a product, the form automatically pre-fills the product name, specs, and default quantities so the user doesn't have to re-type everything. The client catches missing fields before anything sends."

"Once submitted, the user sees estimated delivery times on screen, and the order is already sitting in our spreadsheet. Clean, simple, no friction." `[NEXT SLIDE]`

`SLIDE 09 • DEVELOPER AS ARCHITECT`

The Paradigm Shift: Developer as Architect

"Now let's talk about how we actually wrote the code, because software development right now is completely different from two years ago."

"The old way of learning to code was brutal: sitting there typing out boilerplate line by line, spending three hours digging through Stack Overflow because an npm package broke, and staring at missing semicolons."

"The new way is: developer as architect. You don't waste hours typing boilerplate. You define the system rules, design the data structures, and direct autonomous AI agents to write the code across the files. Your real job is system design and quality control." `[NEXT SLIDE]`

`SLIDE 10 • REALITY CHECK 03`

Reality Check: Learning to Code in 2026

"Reality check number three."

"Teachers and old tutorials always told you: 'Go read 400 pages of documentation before you write a single line of code.'"

"Staying up until 3:00 AM reading manual API docs is completely dead. Today, if you know how to write clear specifications and constraints, you can build full-stack apps in days instead of months." `[NEXT SLIDE]`

`SLIDE 11 • AGENTIC LOOP`

How the Autonomous Coding Loop Works

"And this isn't just asking ChatGPT for code snippets and copy-pasting them like a script kiddie. It's a closed engineering loop with four steps:"

"Step 1: Define the goal. You write a clean spec document, like an AGENTS.md file, detailing your colors, libraries, and rules."

"Step 2: Autonomous edit. The agent reads your repo, figures out what files need changing, and writes the code across multiple files at once."

"Step 3: Automated check. The compiler runs. If there's a build error, the agent reads the terminal output and fixes the bug itself."

"Step 4: Safe history. Every working step gets committed to Git, so if an idea fails, you can roll back in one second." `[NEXT SLIDE]`

`SLIDE 12 • OPENCODE`

Introducing OpenCode

"The tool we recommend for this is OpenCode. It's built by Anomaly, completely open source on GitHub at anomalyco/opencode."

"Unlike web chatbots, it lives right inside your terminal. It has two modes: 'Plan Mode' checks out your files and figures out architecture without touching anything. And 'Build Mode' actually edits the code, runs commands, and builds features."

"You install it on your laptop in 60 seconds with one terminal command: npm install -g @opencode/cli. It runs on Mac, Linux, and Windows." `[NEXT SLIDE]`

`SLIDE 13 • FREE MODELS`

Free AI Models: Zero Cost for Students

"And again: you do not need to spend money on expensive subscriptions."

"Option 1: Free Cloud APIs. Google gives you a huge free tier on Google AI Studio for Gemini 1.5 Flash. Groq Cloud gives you blazing fast Llama 3.3 for zero dollars. And GitHub Models lets students use models like GPT-4o right inside VS Code."

"Option 2: 100% Offline. You can download Ollama on your laptop, run a model like Qwen 2.5 Coder, and code with an AI agent with zero internet connection. Totally private, zero subscription fees." `[NEXT SLIDE]`

`SLIDE 14 • REALITY CHECK 04`

Reality Check: The Compiler Never Lies

"Finally, our last reality check, and this is the most important rule of all. **(small grin)**"

"The AI will happily tell you: 'I fixed all the bugs, everything is perfect!' And then you run the compiler and find 47 errors across three files."

"The golden rule: never trust what an AI claims it wrote. Always let the compiler, the type checker, and your tests be the judge of truth." `[NEXT SLIDE]`

`SLIDE 15 • CONCLUSION & QR`

Thank You & Open Discussion

"To wrap this up: EcoLoop shows that you can tackle real physical material waste and build production-grade web software for zero dollars using modern open tools."

"If you scan the QR code on the screen, it takes you directly to our GitHub repo at github.com/Harman209/ecoloop. You'll find all the source code, the live website catalog, and a one-page setup guide with the exact commands to set up OpenCode and free models on your own machine tonight."

"Thanks for listening, and let's jump into questions." **(done)**